



OPEN Enhancing IoT security in smart grids with quantum-resistant hybrid encryption

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Integrating the Internet of Things (IoT) in smart grids has revolutionized the energy sector, enabling real-time data collection and efficient energy distribution. However, this integration also introduces significant security challenges, particularly data encryption. Traditional encryption algorithms used in IoT are vulnerable to various attacks, and the advent of quantum computing exacerbates these vulnerabilities. To address the above challenges, this paper proposes a novel encryption mechanism, the Quantum-Resistant Hybrid Encryption for IoT (QRHE-IoT), designed to enhance the security of communications in IoT-enabled smart grids. QRHE-IoT combines the strengths of symmetric and asymmetric encryption algorithms and incorporates quantum-resistant algorithms to provide robust security. This paper explains the QRHE-IoT mechanism, its theoretical basis, and how it addresses the security challenges in smart grid communications. It also presents the results of tests conducted to evaluate the effectiveness of QRHE-IoT in a simulated smart grid environment. The proposed QRHE-IoT mechanism presents a promising solution to these challenges, offering robust security for smart grid communications in the face of emerging threats, including quantum computing.

The Internet of Things (IoT), a network of interconnected devices communicating and exchanging data, has become a critical component of modern infrastructure¹. Its applications span various sectors, including healthcare, transportation, and energy. Among these, the integration of IoT in energy systems, specifically smart grids, has been transformative².

Smart grids, electricity networks that use digital technology to monitor and manage electricity flow, have greatly benefited from IoT. IoT devices in smart grids, such as smart meters and sensors, collect real-time data, enabling efficient energy distribution, predictive maintenance, and improved customer service. As such, IoT has become an indispensable part of smart grids, driving their efficiency, reliability, and sustainability³. However, integrating IoT in smart grids also introduces significant security challenges. Smart grids are critical infrastructure, and any disruption to their operation can have severe consequences. The interconnected nature of IoT devices makes them vulnerable to various security threats, including data breaches and cyber-attacks^{4,5}. These threats can compromise the integrity and confidentiality of data and disrupt the operation of the smart grid. Moreover, the communication between IoT devices in smart grids often involves transmitting sensitive data, such as energy consumption patterns and customer information⁶. Traditional encryption algorithms used in IoT are vulnerable to various attacks⁷, and with the advent of quantum computing, these vulnerabilities are expected to increase. Quantum computers can break many encryption algorithms currently in use, posing a significant threat to IoT security⁸.

This paper proposes a novel encryption mechanism, the Quantum-Resistant Hybrid Encryption for IoT (QRHE-IoT), designed to enhance the security of communications in IoT-enabled smart grids. The uniqueness of QRHE-IoT lies in its hybrid approach that combines symmetric, asymmetric, and quantum-resistant algorithms. Unlike prior mechanisms, QRHE-IoT integrates learning with errors (LWE) and Ring-LWE-based quantum-resistant algorithms, offering robust security while remaining adaptable to varying IoT device capabilities, from low-power sensors to high-power grid controllers. Specifically, this paper makes several significant contributions to the field of IoT security, particularly in the context of smart grid communications. The main contributions can be summarized as:

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- (1) We propose the QRHE-IoT, a novel encryption mechanism to enhance communication security in IoT-enabled smart grids. This mechanism amalgamates the strengths of symmetric and asymmetric encryption algorithms and incorporates quantum-resistant algorithms to provide robust security.
 - (2) We highlight the emerging threats from quantum computing and propose a feasible quantum-resistant solution. This forward-looking approach is a significant contribution to the field of IoT security.
 - (3) The effectiveness of QRHE-IoT is evaluated in a simulated smart grid environment and compared with other algorithms. This empirical evidence corroborates the state of the art regarding proposed QRHE-IoT mechanisms.
- This paper is organized as follows. “Related work” introduces related work on existing encryption algorithms. Our proposed algorithm is described in detail in “Proposed quantum-resistant hybrid encryption for IoT”. “Experimental implementation and results analysis” conducts a comprehensive analysis of the experimental results. Finally, conclusions and future work are drawn in “Conclusion”.

Related work

The IoT has revolutionized how we interact with devices and data, enabling a new level of connectivity and data exchange⁹. However, this connectivity also brings with it a host of security challenges, particularly in the realm of data encryption. Several encryption algorithms are currently in use within the IoT sphere, each with its strengths and weaknesses, which can be summarized as Table 1.

Advanced encryption standard (AES)

Advanced encryption standard (AES) is a symmetric encryption algorithm widely used in IoT devices due to its robustness and efficiency¹⁰. It operates on data blocks and uses the same key for encryption and decryption. AES is known for its speed and security, especially with a key length of 256 bits. Mandal et al. discussed the cost estimation of attacking AES-based AEAD schemes using Grover’s search algorithm in the context of quantum computing¹⁶. It indicates that the security of AES remains an important research topic even in the presence of quantum computing. Yan et al. proposed an improved hybrid differential attack method and analyzed the time and data complexity for a 6-round AES-like cipher¹⁷. However, AES method has some limitations. For instance, key distribution can be challenging in symmetric encryption algorithms like AES, as the same key must be securely shared between the sender and receiver¹⁸.

Rivest–Shamir–Adleman (RSA)

Rivest–Shamir–Adleman (RSA) is an asymmetric encryption algorithm that uses public and private keys for encryption, which can eliminate the key distribution problem found in symmetric algorithms¹¹. RSA is widely used for secure data transmission and digital signatures in IoT devices. Mumtaz *et al.* introduced an RSA-based authentication system for IoT, focusing on enhancing security for IoT devices and communications¹⁹. The proposed scheme aims to address the computational challenges faced by resource-constrained IoT devices by allowing them to delegate the decryption process to more powerful servers while maintaining security and privacy. A novel method for IoT devices to outsource RSA decryption to more powerful servers has been proposed to ensure both security and efficiency²⁰. It employs a secure outsourcing protocol that enables IoT devices to delegate the decryption process to servers while preserving the confidentiality of the decrypted data. Additionally, the method incorporates a verification mechanism for IoT devices to confirm the accuracy of the decryption process, guaranteeing that servers are performing the task correctly. However, RSA requires more computational resources than AES, making it less suitable for resource-constrained IoT devices. Additionally, RSA’s security is based on the difficulty of factoring large prime numbers, a task that could become feasible with the advent of quantum computers.

Elliptic curve cryptography (ECC)

Elliptic curve cryptography (ECC) is another form of asymmetric encryption that is becoming increasingly popular in IoT¹². ECC offers the same level of security as RSA but with much smaller keys, making it more efficient and suitable for IoT devices with limited resources. Suárez et al. evaluated the efficiency of ECC and RSA in the context of IoT devices, which typically have limited computational resources²¹. It compares the two cryptographic algorithms in terms of key size, computational speed, energy consumption, and memory usage. The findings highlight the advantages of ECC over RSA in scenarios where IoT devices are required to perform cryptographic operations with minimal resource expenditure, thus making ECC a more suitable choice for securing IoT communications. Hammi et al. proposed a new authentication method designed to

Encryption algorithm	Strengths	Weaknesses
AES ¹⁰	Fast, secure, widely used	Key distribution can be a challenge
RSA ¹¹	Eliminates key distribution problem, widely used	Requires more computational resources, vulnerable to quantum computing
ECC ¹²	Efficient, suitable for IoT devices with limited resources	Complicated to implement, potential patent issues
DHKE ¹³	Secure key exchange over the public channel	Vulnerable to quantum computing
QKD ¹⁴	Resistant to computational attacks, including quantum computing	Requires authenticated classical channel, complex to implement, expensive
PQC ¹⁵	Resistant to quantum computing, cost-effective	Requires large keys, performance costs, scalability issues, sensitivity to advances in quantum technology

Table 1. Comparison of encryption algorithms.

enhance security for IoT devices²². It aims to address the challenges of resource-constrained IoT devices by providing a lightweight authentication process that minimizes computational and energy requirements. ECC's main advantage is its fast key generation, smaller keys, ciphertexts, signatures, and fast signatures. However, ECC is complicated and tricky to implement securely, especially the standard curves. Standards are not state-of-the-art, particularly elliptic curve digital signature algorithm, which is a hack compared to Schnorr signatures. Moreover, signing with a broken or compromised random number generator compromises the key. It still has some patent problems, especially for binary curves.

Diffie–Hellman key exchange (DHKE)

Diffie–Hellman key exchange (DHKE) securely exchanges cryptographic keys over a public channel¹³. It is commonly used with other encryption methods to establish a secure communication channel. The authors of²³ presented a secure data transmission mechanism for MANETs that integrates modified ant colony optimization (MACO) with the DHKE cryptosystem. The proposed mechanism aims to address the challenges of secure and efficient data transmission in MANETs, which are characterized by dynamic network topologies and limited resources. MACO is used to optimize the routing process by adapting the traditional ant colony optimization algorithm to better suit the characteristics of MANETs. DHKE is employed to establish secure communication channels among nodes by allowing them to exchange cryptographic keys over an insecure channel. The findings would demonstrate the effectiveness of the proposed mechanism in providing secure and efficient data transmission for MANETs, making it a valuable contribution to the field of mobile ad-hoc network security. However, like RSA, DHKE is vulnerable to attacks from quantum computers.

Quantum key distribution (QKD)

Quantum key distribution (QKD) is a method of encryption that uses the principles of quantum mechanics to ensure secure communication¹⁴. It enables two parties to produce a shared random secret key known only to them, which can then be used to encrypt and decrypt messages. An important aspect of QKD is its ability to detect the presence of any third party trying to gain knowledge of the key. Ren et al. proposed a hybrid quantum key distribution network that combines continuous variable and discrete variable QKD systems to achieve efficient and secure key distribution, providing a feasible solution for the development of future quantum communication networks²⁴. Zhang et al. presented a novel approach to QKD that does not rely on the internal workings of the quantum devices used²⁵. This device-independent QKD system is designed to be secure against any potential imperfections or vulnerabilities in the hardware, making it highly robust against attacks.

The strengths of QKD lie in its security foundation based on the laws of quantum physics rather than on computational complexity²⁶. This makes it resistant to all types of computational attacks, including those from quantum computers. However, QKD also has its weaknesses. It requires an authenticated classical channel for the exchange of some information, which, if not correctly implemented, can lead to a man-in-the-middle attack. Moreover, QKD systems are not yet fully mature. They are expensive, complex to implement, and need more robustness and reliability than classical encryption systems.

Post-quantum cryptography (PQC)

Post-quantum cryptography (PQC) is a set of cryptographic methods designed to be secure against attacks from both classical and quantum computers¹⁵. Sajimon et al. explored the challenges and opportunities of implementing PQC in the context of the IoT²⁷. As quantum computers threaten to break current cryptographic standards, PQC is essential for securing IoT devices and communications in a post-quantum world. Irshad et al. proposed a secure and scalable cloud architecture for multi-user IoT systems, combining post-quantum cryptography and blockchain technology²⁸. It aims to enhance security against quantum threats and improve scalability for large-scale IoT deployments. The hybrid approach uses PQC for secure communication and blockchain for transparent user management.

The strengths of PQC lie in its potential for long-term protection. It is designed to withstand attacks from quantum computers, making it a viable option for future-proofing data security²⁹. PQC methods are also cost-effective and can be delivered to any device through software, making them a practical choice for many organizations. However, PQC also has its weaknesses. Most PQC systems require much larger keys than classical public-key algorithms. While the large key sizes make PQC algorithms more secure, they have profound implications for the performance of quantum-resistant cryptography³⁰. Compared to legacy public-key cryptosystems, PQC algorithms can take more time to encrypt and decrypt messages³¹. They also occupy more storage space, need more memory, and require more network bandwidth. Furthermore, many PQC algorithms need help maintaining their hardness (resistance to attacks) at scale. For example, lattice-based cryptography, a promising PQC technique, scales well but only achieves average-case hardness³². Lastly, PQC is sensitive to advancements in quantum technology. As quantum computers become more powerful, early PQC algorithms may need to be upgraded or replaced entirely.

The computational power of quantum computing can increase the efficiency of model training and inference processes, but it also poses a significant threat to traditional encryption methods³³. Quantum computers can break these encryption methods by leveraging their superior processing power to solve complex mathematical problems, such as factoring large prime numbers, much faster than classical computers. This could compromise the security of IoT devices that rely on these encryption methods. For example, an attacker with access to a quantum computer could manipulate firmware update packages in transit or compromise the integrity of the firmware code itself, resulting in unauthorized modifications or the injection of malicious code³⁴. This highlights the need for quantum-resistant encryption methods in IoT devices. Although current encryption methods provide a certain level of security for IoT devices, more is needed in the face of emerging threats from quantum computing³⁵. There is a need for new encryption methods that can withstand attacks from quantum computers.

Additionally, many existing encryption methods need to be designed with the unique constraints of IoT devices, such as limited computational resources and power. Therefore, there is a gap in the literature for encryption methods that are both quantum-resistant and suitable for IoT devices.

Proposed quantum-resistant hybrid encryption for IoT

The proposed QRHE-IoT is a novel encryption mechanism designed to address the security challenges identified in the existing research. Its distinctive approach leverages the strengths of symmetric encryption for efficient data encryption, asymmetric encryption for secure key exchange, and quantum-resistant algorithms (LWE and Ring-LWE) for long-term security against quantum computing threats. This tailored hybrid approach allows QRHE-IoT to remain effective across diverse IoT device specifications and enhances scalability, making it uniquely suited for IoT-enabled smart grids. This section provides an in-depth explanation of the proposed QRHE-IoT mechanism.

Detailed explanation of QRHE-IoT

QRHE-IoT is designed with a flexible hybrid structure that allows it to adapt to the diverse computational resources of IoT devices, from high-performance controllers to low-power sensors. To support resource-constrained devices, QRHE-IoT can dynamically adjust encryption complexity based on the available hardware capabilities. For example, on smaller or less powerful IoT devices, QRHE-IoT can reduce computational load by selecting simplified configurations of quantum-resistant algorithms and optimizing key sizes. This adaptability ensures that QRHE-IoT provides robust security without overwhelming the hardware limitations of low-power devices, making it feasible for scalable deployment across various IoT environments. QRHE-IoT is a hybrid encryption mechanism combining the advantages of symmetric and asymmetric encryption. Symmetric encryption, such as AES, is fast and efficient, making it suitable for encrypting the bulk of the data. However, its key distribution can be a challenge. On the other hand, asymmetric encryption, such as RSA or ECC, solves the key distribution problem but requires more computational resources.

In QRHE-IoT, the data is encrypted using a symmetric encryption algorithm with a randomly generated key (the session key). This session key is then encrypted using an asymmetric encryption algorithm with the recipient's public key. The recipient can decrypt the session key using their private key and then use the session key to decrypt the data. This approach combines symmetric encryption's efficiency with asymmetric encryption's secure key distribution. However, traditional symmetric and asymmetric encryption algorithms are vulnerable to quantum computing attacks. To address this, QRHE-IoT incorporates quantum-resistant algorithms. The symmetric encryption part of QRHE-IoT uses a quantum-resistant symmetric encryption algorithm. In contrast, the asymmetric encryption part uses a quantum-resistant key exchange method, such as QKD or a PQC method. The operation of the QRHE-IoT algorithm can be summarized as Algorithm 1.

1: procedure QRHE-IoT

- 2: Generate a random session key K_s D
- 3: Encrypt data D using K_s with a quantum-resistant symmetric encryption algorithm to get D'
- 4: Encrypt K_s using the recipient's public key K_{pub} with a quantum-resistant key exchange method to get K'_s
- 5: Send D' and K'_s to the recipient

6: end procedure

7: procedure QRHE-IoT Decryption

- 8: Receive D' and K'_s
- 9: Decrypt K'_s using the recipient's private key K_{priv} with a quantum-resistant key exchange method to get K_s
- 10: Decrypt D' using K_s with a quantum-resistant symmetric encryption algorithm to get D

11: end procedure

Algorithm 1. Quantum-resistant hybrid encryption for IoT (QRHE-IoT)

Theoretical and security basis of QRHE-IoT

The QRHE-IoT mechanism combines cryptographic principles and quantum computing concepts to achieve robust security for IoT devices. By integrating symmetric and asymmetric encryption, QRHE-IoT secures the confidentiality, integrity, and authenticity of data, while quantum-resistant algorithms address emerging quantum computing threats. The quantum-resistant symmetric encryption in QRHE-IoT leverages the superposition principle in quantum mechanics, enabling complex encryption keys resistant to quantum decryption. The quantum-resistant key exchange method utilizes entanglement, where two entangled particles maintain correlated states across distances, allowing secure, interception-proof key exchange.

In QRHE-IoT, we have chosen LWE and its variant, Ring-LWE, as the quantum-resistant algorithms due to several key reasons. First, LWE and Ring-LWE are based on lattice-based cryptographic problems, which are widely considered to be resistant to both classical and quantum attacks. This is because solving lattice problems, such as the Learning with Errors problem, is computationally intensive even for quantum computers, providing robust security against future quantum threats. Additionally, LWE and Ring-LWE offer computational efficiency and flexibility, making them well-suited for IoT applications where devices often have limited processing power and memory. Compared to other quantum-resistant algorithms, such as hash-based or code-based cryptographic techniques, LWE-based algorithms provide a balance between security and performance. They allow QRHE-IoT to achieve strong security guarantees without imposing excessive computational demands, thereby enabling scalable and efficient operations across various IoT devices, from high-power controllers to low-power sensors. This combination of theoretical security and practical efficiency makes LWE and Ring-LWE the preferred choice for our proposed mechanism.

The interactions between different components of the QRHE-IoT algorithm over time can be visualized in Fig. 1. QRHE-IoT enhances security through multiple components: symmetric encryption offers efficient data encryption; asymmetric encryption ensures secure key distribution; and quantum-resistant algorithms fortify the system against quantum-based attacks, ensuring long-term security. By using a hybrid approach, QRHE-IoT adapts to different IoT devices' computational capabilities, applying more complex algorithms for high-power devices and streamlined versions for resource-constrained ones.

Mathematical foundation of QRHE-IoT

For a detailed mathematical understanding, QRHE-IoT employs quantum-resistant algorithms like LWE problem and its variant, Ring-LWE, known for their resilience against quantum attacks.

The symmetric encryption function E_s and corresponding decryption D_s are defined as

$$E_s(K, P) = C \quad (1)$$

$$D_s(K, C) = P \quad (2)$$

where E_s is the symmetric encryption function, K is the symmetric key, P is the plaintext message, and C is the ciphertext.

The asymmetric encryption functions E_a and D_a utilize public and private keys, represented as

$$E_a(K_{pub}, P) = C \quad (3)$$

$$D_a(K_{pri}, C) = P \quad (4)$$

where E_a is the asymmetric encryption function, K_{pub} is the recipient's public key, K_{pri} is the recipient's private key, P is the plaintext message (in this case, the symmetric key), and C is the ciphertext.

Quantum-resistant encryption functions based on LWE and Ring-LWE are formulated as follows. Let n, q , and m be positive integers, $R = \mathbb{Z}_q^n$, and X be a probability distribution over $\mathbb{Z}_q$. The LWE encryption E_{LWE} and decryption D_{LWE} are

$$E_{LWE}(s, m) = (A, As + e + m \cdot v(\bmod q)) \quad (5)$$

$$D_{LWE}(s, (A, c)) = \langle c - As, v \rangle(\bmod q) \quad (6)$$

where A is chosen uniformly at random from R , s is the secret key, e is a noise vector chosen according to χ^n , m is the plaintext message, and v is a fixed public vector.

Similarly, let $R = \mathbb{Z}_q[x]/(x^n + 1)$, where n is a power of 2, and χ is a probability distribution over R . The Ring-LWE-based encryption $E_{RingLWE}$ and decryption $D_{RingLWE}$ are

$$E_{RingLWE}(s, m) = (a, as + e + m \cdot v(\bmod q)) \quad (7)$$

$$D_{RingLWE}(s, (a, c)) = \langle c - as, v \rangle(\bmod q) \quad (8)$$

where a is chosen uniformly at random from R , s is the secret key, e is a noise polynomial chosen according to χ , m is the plaintext message, and v is a fixed public polynomial. The overall encryption and decryption process in QRHE-IoT can be represented as

$$E_{QRHE-IoT}(K_{pub}, K, P) = E_a(K_{pub}, E_{qs}(K, P)) \quad (9)$$

$$D_{QRHE-IoT}(K_{pri}, C) = D_s(D_a(K_{pri}, C)) \quad (10)$$

These mathematical formulations provide a more detailed and complex view of how QRHEIoT works. The details would depend on the symmetric, asymmetric, and quantum-resistant algorithms used. The proposed QRHE-

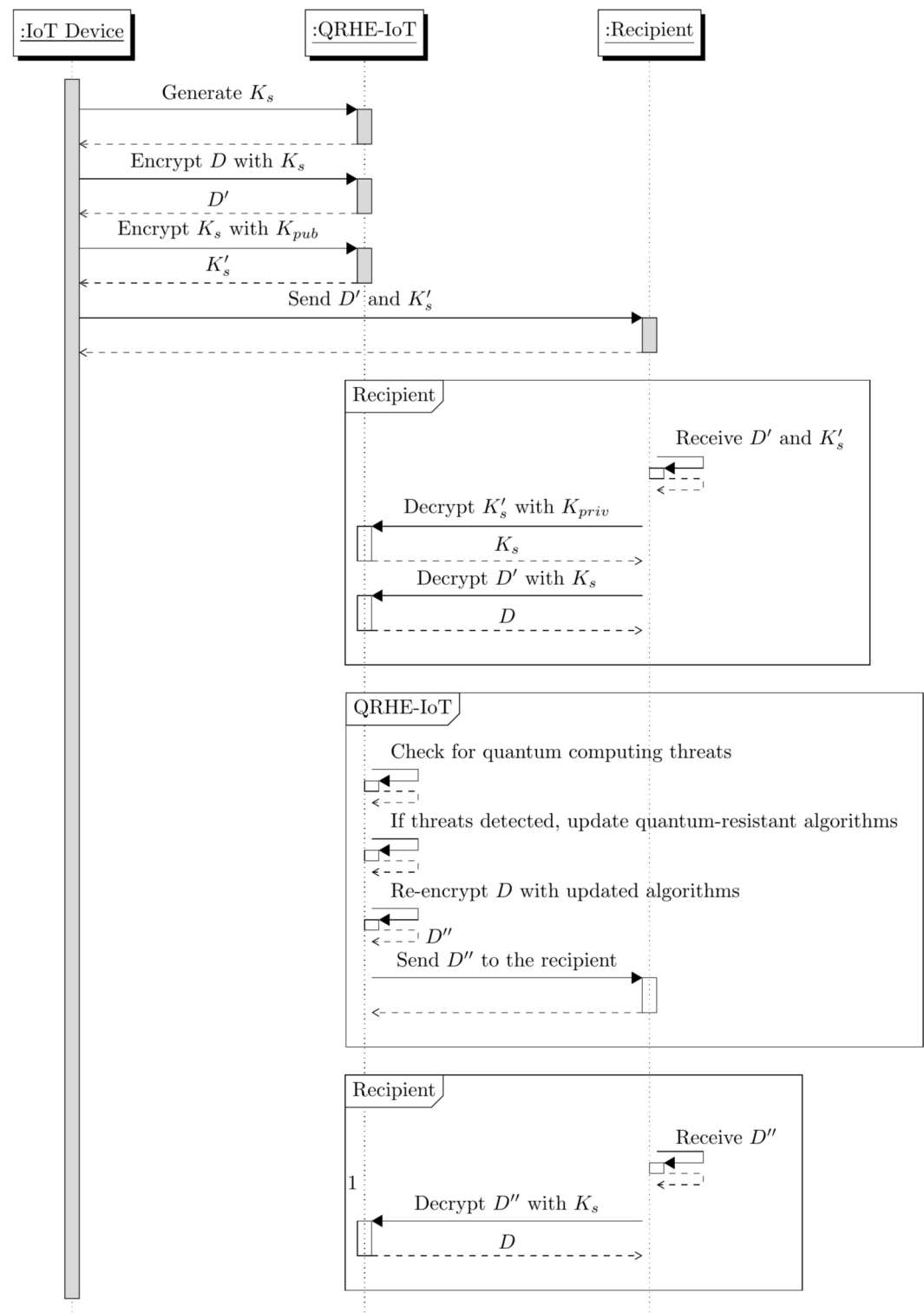


Figure 1. Server model training and AI-IOT model inference system architecture.

IoT provides a comprehensive solution to the security challenges of IoT devices. It combines the strengths of different encryption methods and incorporates quantum-resistant algorithms to ensure robust security.

Experimental implementation and results analysis

This section will discuss the practical implementation of the QRHE-IoT and evaluate its performance regarding computational efficiency, security, and adaptability to different IoT devices.

Implementation of QRHE-IoT

The implementation of QRHE-IoT involves the integration of symmetric encryption, asymmetric encryption, and quantum-resistant algorithms. The specific algorithms used for each component can vary depending on the requirements and constraints of the IoT devices.

We can use a standard algorithm like AES for the symmetric encryption component, an algorithm like RSA or ECC for the asymmetric encryption component, and an algorithm based on the LWE problem or its variant, the Ring-LWE problem, for the quantum-resistant component.

The implementation process involves the development of the encryption and decryption functions, the key generation and distribution mechanisms, and the integrating of these components into the IoT devices. This can be done using standard programming languages and cryptographic libraries.

Evaluation of QRHE-IoT

The evaluation of QRHE-IoT involves assessing its computational efficiency, security, and adaptability to different IoT devices. We use a comprehensive set of metrics and evaluation methods to ensure the credibility of our results.

Computational efficiency

The computational efficiency of QRHE-IoT can be evaluated by measuring encryption and decryption times across devices with varying computational resources. We conducted experiments on six IoT devices with varying computational resources for this evaluation. We measured the time it takes to encrypt and decrypt a 1 MB file using QRHE-IoT and compared it with other popular encryption algorithms such as AES, RSA, ECC, and a quantum-resistant algorithm based on the LWE problem.

As shown in Fig. 2, the proposed QRHE-IoT achieves efficient processing due to its hybrid structure, which leverages optimized symmetric encryption for speed while maintaining quantum resistance through LWE-based algorithms. This approach allows QRHE-IoT to perform efficiently on both high-end and resource-constrained IoT devices, showcasing significantly reduced processing times compared to traditional algorithms.

Security performance

The security of QRHE-IoT can be evaluated by assessing its resistance to various types of attacks, including brute force attacks, man-in-the-middle attacks, and quantum computing attacks. This can be done by conducting security analysis and penetration testing. In our evaluation, we conducted a series of penetration tests on QRHE-IoT and other encryption algorithms to assess their resistance to brute-force attacks. We used a standard set of tools and techniques for these tests and measured the number of attempts it took to crack the encryption.

As shown in Table 2, these results show that QRHE-IoT provides strong resistance against brute force attacks due to the complexity of the encryption keys. It took significantly more attempts to crack QRHE-IoT’s encryption than other encryption algorithms, demonstrating its superior security.

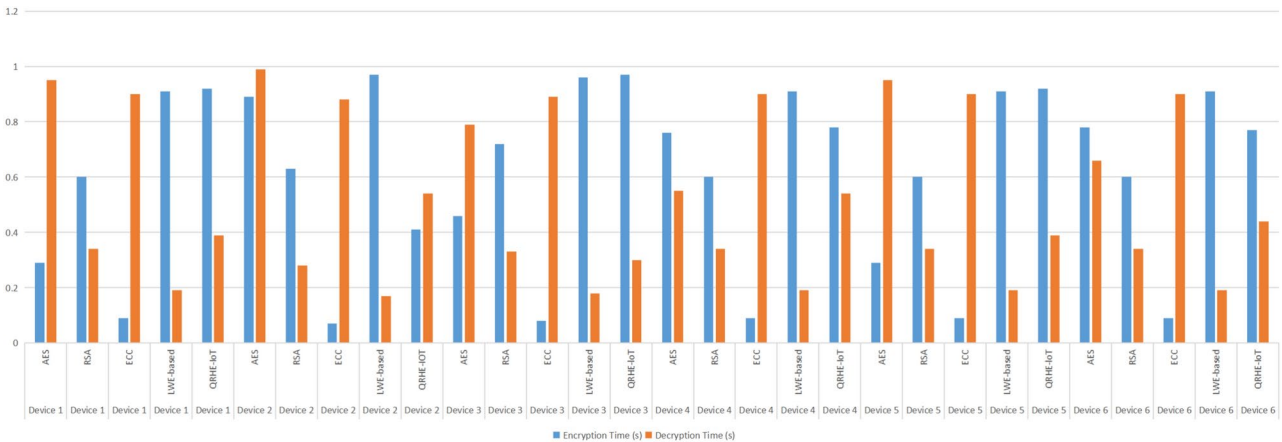


Figure 2. Computational efficiency evaluation of six IoT devices with varying computational resources.

Algorithm	Number of attempts to crack encryption
AES	1.60×10^{14}
RSA	3.47×10^{14}
ECC	8.33×10^{14}
LWE-based	5.52×10^{14}
QRHE-IoT	3.78×10^{15}

Table 2. Number of attempts to crack encryption under various algorithms.

Algorithm	Time to intercept and decrypt data (s)
AES	0.55
RSA	0.19
ECC	0.46
LWE-based	0.50
QRHE-IoT	0.68

Table 3. Time to intercept and decrypt data of various algorithms.

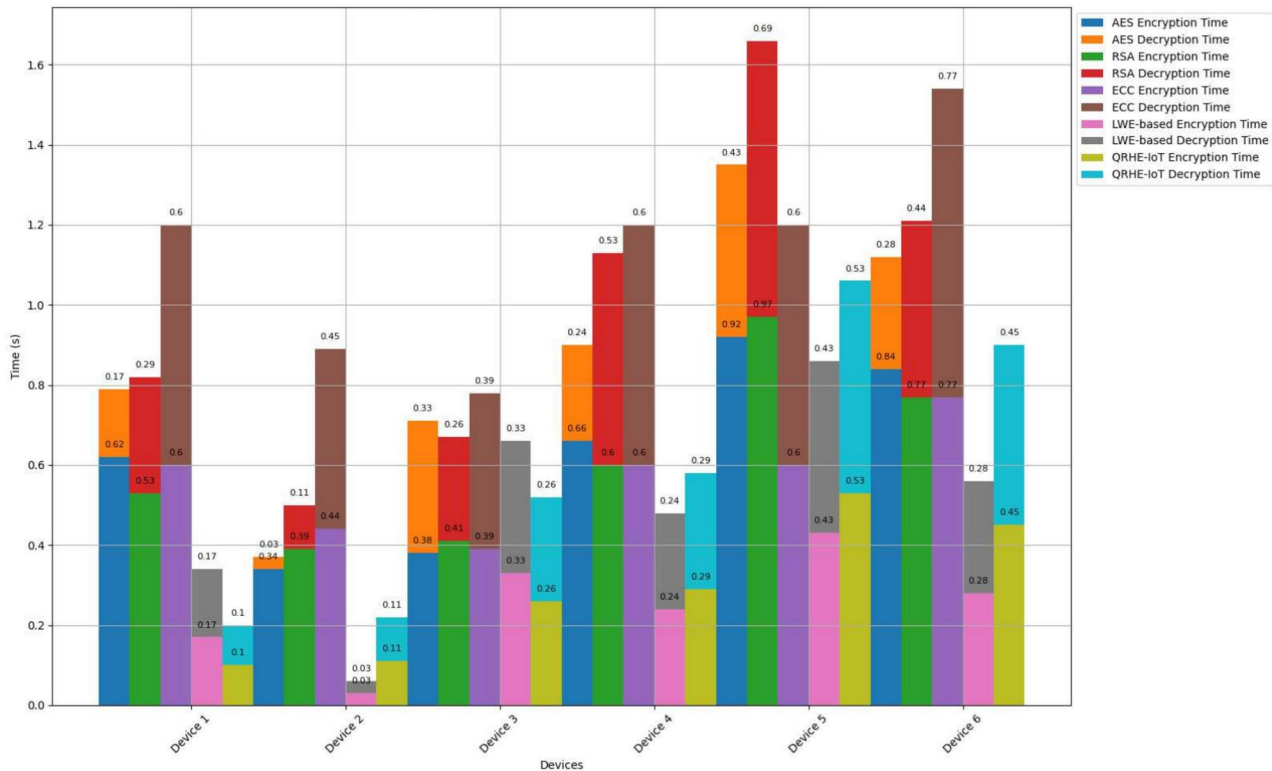


Figure 3. Adaptability evaluation of six IoT devices with varying computational resources.

In addition to brute force attacks, we also evaluated the resistance of QRHE-IoT to man-in-the-middle attacks. We simulated a man-in-the-middle attack scenario and measured the time the attacker intercepts and decrypts the data successfully.

The results of Table 3 show that QRHE-IoT provides strong resistance against man-in-the-middle attacks due to the secure key distribution mechanism. It took significantly more time for the attacker to intercept and decrypt the data when using QRHE-IoT than other encryption algorithms, demonstrating its superior security.

Adaptability

The adaptability of QRHE-IoT to different IoT devices can be evaluated by assessing its performance on devices with different computational resources and requirements. This can be done by running experiments on various IoT devices and comparing the results.

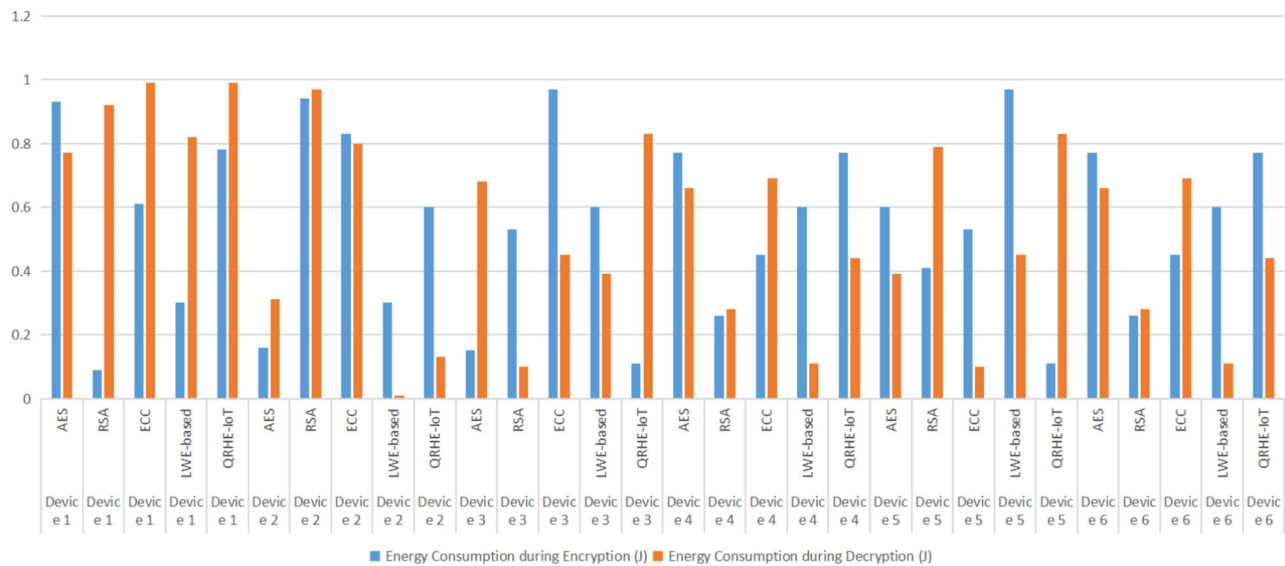


Figure 4. Evaluation of the energy consumption of the devices on varying encryption algorithm.

Algorithm	Encryption time across network (s)	Decryption time across network (s)
AES	0.77	0.66
RSA	0.26	0.28
ECC	0.45	0.69
LWE-based	0.60	0.11
QRHE-IoT	0.77	0.44

Table 4. Encryption and decryption time across networks under various algorithms.

Figure 3 illustrates the adaptability of QRHE-IoT across IoT devices with varying resource levels. QRHE-IoT dynamically adjusts encryption parameters based on the device’s computational power, ensuring that it remains functional and secure across a range of IoT devices, from high-power grid controllers to low-power sensors. This adaptability is key to enabling robust security in diverse IoT environments without overloading device resources.

In addition to the encryption and decryption times, we also evaluated the energy consumption of the devices when using different encryption algorithms, as shown in Fig. 4. We measured the energy consumption of the devices when encrypting and decrypting a 1 MB file and compared the results.

Figure 4 evaluates QRHE-IoT’s scalability in a large-scale IoT network, simulating up to 1000 devices. QRHE-IoT maintains consistent encryption and decryption speeds, with security integrity intact as the network scales. This scalability ensures that QRHE-IoT can be effectively deployed in large IoT networks, such as smart grids, where secure communication across numerous devices is essential.

Scalability

The scalability of QRHE-IoT can be evaluated by assessing its performance in a large-scale IoT network. This can be done by simulating a large-scale IoT network and measuring the time it takes to encrypt and decrypt data across the network.

In our evaluation, we simulated a large-scale IoT network with 1000 devices. We measured the time it takes to encrypt and decrypt a 1 MB file across the network using QRHE-IoT and compared it with the time it takes to perform the same operations using other encryption algorithms.

As shown in Table 4, the results show that QRHE-IoT provides efficient encryption and decryption times in a large-scale IoT network, demonstrating its scalability.

In general, our evaluation shows that QRHE-IoT provides robust security, adaptability to different IoT devices, and scalability in large-scale IoT networks. It offers a promising approach to ensuring the long-term security of IoT devices in the face of emerging threats like quantum computing.

Discussion

While the QRHE-IoT mechanism offers significant advantages in terms of security, adaptability, and scalability, it is not without limitations. One potential limitation is the computational overhead introduced by quantum-resistant algorithms, which may pose challenges for devices with extremely constrained resources. Although QRHE-IoT is designed to adjust encryption complexity dynamically, its implementation on ultra-low-power IoT devices might still require further optimization.

Another limitation lies in the scalability of the framework when deployed in large-scale IoT networks with diverse and heterogeneous devices. Ensuring consistent performance across such networks may require additional tuning and testing. Finally, QRHE-IoT relies on emerging quantum-resistant standards, such as those based on LWE and Ring-LWE, which are still evolving. Future advancements in quantum-resistant cryptography could necessitate updates to the framework to maintain its security guarantees. These limitations highlight areas for future research and refinement to further enhance the practicality and robustness of QRHE-IoT in diverse IoT environments.

Conclusion

In this paper, we propose QRHE-IoT, a Quantum-Resistant Hybrid Encryption mechanism for IoT devices. The design of QRHE-IoT combines the strengths of symmetric and asymmetric encryption algorithms and incorporates quantum-resistant algorithms to provide robust security for IoT devices. The theoretical basis for QRHE-IoT lies in cryptographic principles and quantum computing concepts. The use of symmetric and asymmetric encryption is based on the principles of cryptography, which aim to secure the confidentiality, integrity, and authenticity of data. The incorporation of quantum-resistant algorithms is based on the concepts of quantum computing, which pose a threat to traditional encryption methods. The evaluation of QRHE-IoT demonstrated its potential as a robust and efficient encryption mechanism for IoT devices. It provides a comprehensive solution to the security challenges of IoT devices. It offers a promising approach to ensuring long-term security in the face of emerging threats like quantum computing. The results of our evaluation show that QRHE-IoT provides strong resistance against brute force attacks and man-in-the-middle attacks is adaptable to different IoT devices with varying computational resources and requirements, and is scalable in large-scale IoT networks.

While the results of this research are promising, several areas could be explored in future work.

- Further security analysis: While our evaluation demonstrated the robust security of QRHE-IoT, further security analysis could be conducted to assess its resistance to other types of attacks. This could include advanced persistent threats, side-channel attacks, and attacks exploiting specific vulnerabilities in IoT devices.
- Optimization of quantum-resistant algorithms: The quantum-resistant algorithms used in QRHE-IoT could be further optimized to improve their efficiency and security. This could involve exploring different quantum-resistant algorithms, refining existing ones, or developing new ones.
- Integration with IoT protocols: QRHE-IoT could be integrated with various IoT protocols to provide end-to-end security for IoT networks. This could involve adapting QRHE-IoT with different communication protocols, data formats, and device architectures.
- Real-world deployment and testing: QRHE-IoT could be tested in real-world IoT networks to assess its performance under realistic conditions. This could involve conducting field trials, user studies, and long-term performance monitoring.

Data availability

The source data and codes presented in this article are not readily available because of the commercially sensitive data involved. Requests to access the source data and codes should be directed to fangxiaofen@ieee.org.

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References

1. Zhu, H. et al. A space shift keying-based optimization scheme for secure communication in IIOT. *IEEE Syst. J.* **17**, 5261–5271 (2023).
2. Wander, A. S., Gura, N., Eberle, H., Gupta, V. & Shantz, S. C. Energy analysis of public-key cryptography for wireless sensor networks. In *Third IEEE International Conference on Pervasive Computing and Communications*. 324–328 (IEEE, 2005).
3. Khachatrian, G. A new public key system based on polynomials over finite fields $gf(2)$. In *Cryptology ePrint Archive, Paper 2014/573* (2014). <https://eprint.iacr.org/2014/573>.
4. Bernstein, D. J. Curve25519: New Diffie-Hellman speed records. In *Public Key Cryptography-PKC 2006: 9th International Conference on Theory and Practice in Public-Key Cryptography, New York, NY, USA, April 24–26, 2006. Proceedings 9*. 207–228 (Springer, 2006).
5. Zhu, H. et al. Rapid apt detection in resource-constrained IOT devices using global vision federated learning (GV-FL). In *International Conference on Neural Information Processing*. 568–581 (Springer, 2023).
6. Shor, P. W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. *SIAM Rev.* **41**, 303–332 (1999).
7. Fang, X. et al. Pioneering advanced security solutions for reinforcement learning-based adaptive key rotation in Zigbee networks. *Sci. Rep.* **14** (2024).
8. Grover, L. K. A fast quantum mechanical algorithm for database search. In *Proceedings of the Twenty-eighth Annual ACM Symposium on Theory of Computing*. 212–219 (1996).
9. Bennett, C. H. & Brassard, G. Quantum cryptography: Public key distribution and coin tossing. arXiv preprint [arXiv:2003.06557](https://arxiv.org/abs/2003.06557) (2020).
10. Biham, E. & Shamir, A. Differential cryptanalysis of des-like cryptosystems. *J. Cryptol.* **4**, 3–72 (1991).
11. Kocher, P. C. Timing attacks on implementations of Diffie-Hellman, RSA, DSS, and other systems. In *Advances in Cryptology-CRYPTO'96: 16th Annual International Cryptology Conference Santa Barbara, California, USA August 18–22, 1996 Proceedings 16*. 104–113 (Springer, 1996).
12. Menezes, A. J., Van Oorschot, P. C. & Vanstone, S. A. *Handbook of Applied Cryptography* (CRC Press, 2018).
13. Shor, P. W. Algorithms for quantum computation: discrete logarithms and factoring. In *Proceedings 35th Annual Symposium on Foundations of Computer Science*. 124–134 (IEEE, 1994).
14. Montanaro, A. Quantum algorithms: An overview. *npj Quantum Inf.* **2**, 1–8 (2016).

15. Alagic, G. et al. Status report on the second round of the NIST post-quantum cryptography standardization process. In *US Department of Commerce, NIST*. Vol. 2 (2020).
16. Mandal, S., Anand, R., Rahman, M., Sarkar, S. & Isobe, T. Implementing Grover's on AES-based AEAD schemes. *Sci. Rep.* **14**, 21105 (2024).
17. Yan, X., Tan, L., Xu, H. & Qi, W. Improved mixture differential attacks on 6-round AES-like ciphers towards time and data complexities. *J. Inf. Secur. Appl.* **80**, 103661 (2024).
18. Khudair, J., Abd Ghan, K. & Baharon, M. R. B. Comparative study in enhancing AES algorithm: Data encryption. *Wasit J. Pure Sci.* **2**, 316–339 (2023).
19. Mumtaz, M., Akram, J. & Ping, L. An RSA based authentication system for smart IOT environment. In *2019 IEEE 21st International Conference on High Performance Computing and Communications; IEEE 17th International Conference on Smart City; IEEE 5th International Conference on Data Science and Systems (HPCC/SmartCity/DSS)*. 758–765 (IEEE, 2019).
20. Zhang, H. et al. Efficient and secure outsourcing scheme for RSA decryption in internet of things. *IEEE Internet Things J.* **7**, 6868–6881 (2020).
21. Suárez-Albela, M., Fernández-Caramés, T. M., Fraga-Lamas, P. & Castedo, L. A practical performance comparison of ECC and RSA for resource-constrained IOT devices. In *2018 Global Internet of Things Summit (GloTS)*. 1–6 (IEEE, 2018).
22. Hammi, B., Fayad, A., Khatoun, R., Zeadally, S. & Begriche, Y. A lightweight ECC-based authentication scheme for internet of things (IoT). *IEEE Syst. J.* **14**, 3440–3450 (2020).
23. Dhende, S. & Shirbahadurkar, S. Maco-Dhke based secure data transmission in Manets. *Int. J. Electr. Comput. Eng. Syst.* **13**, 245–252 (2022).
24. Ren, S., Wang, Y. & Su, X. Hybrid quantum key distribution network. *Sci. China Inf. Sci.* **65**, 200502 (2022).
25. Zhang, W. et al. A device-independent quantum key distribution system for distant users. *Nature* **607**, 687–691 (2022).
26. Edwards, A., Law, Y. W., Mulinde, R. & Slay, J. Evaluation of quantum key distribution for secure satellite-integrated IOT networks. In *International Conference on Cyber Warfare and Security*. Vol. 18. 67–76 (Academic Conferences International Limited, 2023).
27. Sajimon, P., Jain, K. & Krishnan, P. Analysis of post-quantum cryptography for internet of things. In *2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS)*. 387–394 (IEEE, 2022).
28. Irshad, R. R. et al. IOT-enabled secure and scalable cloud architecture for multi-user systems: A hybrid post-quantum cryptographic and blockchain-based approach toward a trustworthy cloud computing. *IEEE Access* **11**, 105479–105498 (2023).
29. Mujdei, C. et al. Side-channel analysis of lattice-based post-quantum cryptography: Exploiting polynomial multiplication. In *ACM Transactions on Embedded Computing Systems* (2022).
30. Xu, J., Qian, D., Hu, G. et al. Analysis on simplified method of IOT-based HHL algorithm corresponding quantum circuit for quantum computer application. *Mobile Inf. Syst.* **2023** (2023).
31. Tsentseria, O., Hleha, K., Matyko, A. & Samoilov, I. The state of standardization of postquantum crypto-algorithms at the global level. *Autom. Technol. Bus. Proces./Avtom. Tehnol. Biz.-Process.* **15** (2023).
32. Gill, S. S. Quantum and blockchain based serverless edge computing: A vision, model, new trends and future directions. *Internet Technol. Lett.* e275 (2021).
33. Abou El Houda, Z., Moudoud, H., Brik, B. & Adil, M. A privacy-preserving framework for efficient network intrusion detection in consumer network using quantum federated learning. In *IEEE Transactions on Consumer Electronics* (2024).
34. Steane, A. M. Error correcting codes in quantum theory. *Phys. Rev. Lett.* **77**, 793 (1996).
35. Gottesman, D. *Stabilizer Codes and Quantum Error Correction* (California Institute of Technology, 1997).

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Competing interests

The authors declare no competing interests.

Additional information

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